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DATA PRIVACY AND AI CONCEPTS

Introduction to Data Privacy

What is data privacy?

- ◆ Data privacy refers to the proper handling, processing, storage, and usage of personal information to protect individuals' rights and freedoms.
- ◆ It is about protecting personal data from being misused, leaked, or accessed without permission.

Examples: Your birthday, phone number, and photos are private unless you choose to share them.

Difference Between Security and Privacy.

- ◆ **Security:** Refers to the measures and practices put in place to protect data and systems from unauthorized access, attacks, or damage. It encompasses a wide range of technologies.
- ◆ **Privacy:** Refers to the rights and expectations of individuals regarding the collection, use, and sharing of their personal information.
- ◆ Security focuses on protecting data from threats, while privacy focuses on the data handling

Regulations

GDPR (General Data Protection Regulation-EU)

- ◆ GDPR establishes comprehensive data protection and privacy regulations for individuals within the European Union (EU). It sets guidelines for the collection, processing, storage, and sharing of personal data.
- ◆ It grants individuals rights over their personal data, including the right to access, rectify, erase, and restrict processing of their data.

Used to:

- ◆ Protect Individual Privacy
- ◆ Standardize Data Protection
- ◆ Enhance Trust

CCPA (California Consumer Privacy Act)

- ◆ CCPA enhances privacy rights and consumer protection for residents of California. It requires businesses to disclose their data collection practices and gives consumers the right to know, access, and delete their personal information.
- ◆ It also allows consumers to opt-out of the sale of their personal data

Key Rights:

- ◆ Right to know what personal info is collected and why.
- ◆ Right to delete personal data.
- ◆ Right to opt-out of the sale of their personal information.
- ◆ Right to non-discrimination in exercising privacy rights.

HIPAA (Health Insurance Portability and Accountability Act -USA)

- ◆ HIPAA establishes national standards for the protection of health information in the U.S. It governs how healthcare providers, health plans, and other entities handle protected health information (PHI).
- ◆ It includes provisions for the privacy and security of PHI, as well as breach notification requirements.

Used to

- Protect Patient Privacy
- Standardize Health Information Management
- Enhance Trust in Healthcare

KEY DATA PRIVACY TECHNIQUES

Anonymization:

Anonymization is the process of making personal data unidentifiable by removing names, contact info, IDs, or anything that can reveal a person's identity.

Uses:

- ◆ To protect people's privacy
- ◆ To use data safely in research, AI training, statistics, etc.
- ◆ To comply with laws like GDPR or HIPAA

Pseudonymization:

Pseudonymization is a data processing technique that replaces or removes identifiable information from a dataset, allowing the data to be processed without directly revealing the identity of the individuals involved. This technique is often used to enhance privacy and data protection while still enabling data analysis and processing.

Example: Changing "John" to "User1234" while keeping track of their actions.

Used in:

- ◆ Analytics
- ◆ Marketing
- ◆ Partial data protection

Tokenization:

Tokenization is a data security technique that involves replacing sensitive data elements with non-sensitive equivalents, known as tokens. These tokens retain essential information about the original data but have no extrinsic value or meaning outside of the specific context in which they are used. Tokenization is commonly used to protect sensitive information, such as credit card numbers, personal identification numbers (PINs), and other confidential data.

Used in:

- ◆ Payment processing
- ◆ Data security
- ◆ Compliance
- ◆ Data Analytics

Data Masking:

Data Masking is a data protection technique that involves obscuring or altering sensitive data to prevent unauthorized access while maintaining its usability for testing, development, or analysis purposes. The goal of data masking is to protect sensitive information, such as personally identifiable information (PII), financial data, or health records, while still allowing organizations to work with realistic data sets.

Uses:

- ◆ Testing and Development
- ◆ Data Sharing
- ◆ Compliance
- ◆ Training

Encryption:

Encryption is a process that transforms readable data (plain text) into an unreadable format (cipher text) to protect it from unauthorized access. It is a fundamental aspect of data security, ensuring that sensitive information remains confidential. There are two primary types of encryption symmetric encryption and asymmetric encryption.

AI CONCEPTS RELATED TO DATA PRIVACY

What is AI hallucination?

AI hallucination refers to instances when artificial intelligence models generate information that is incorrect, misleading, or entirely fabricated, even though it appears plausible. This phenomenon occurs due to various factors, including limitations in training data and the model's design, leading to outputs that do not accurately reflect reality.

Examples: Fabricated Information, Incorrect Predictions

Prompt Injection Attacks

Prompt injection attacks are a type of security threat in AI where malicious users manipulate input prompts to deceive generative AI systems, causing them to produce unintended or harmful outputs. This exploitation can bypass the AI's built-in safety measures and intended instructions, leading to significant vulnerabilities.

Data Leakage in LLMs

Data leakage in large language models (LLMs) refers to the unintended exposure of sensitive or confidential information during the model's operation. This can occur when the model reveals proprietary data or personal information that was part of its training dataset, leading to privacy breaches and security risks.

Differential Privacy

Differential privacy is a mathematical framework designed to protect individual privacy when analyzing datasets. It ensures that the inclusion or exclusion of a single individual's data does not significantly affect the outcome of any analysis, thereby safeguarding personal information while still allowing for useful insights.

ZOHO CREATOR - Overview

Zoho Creator is a low-code application development platform that allows users to build custom apps without deep programming knowledge. It provides a drag-and-drop interface to create forms, reports, and workflows quickly and efficiently. With built-in automation and scripting (using Deluge), users can manage complex logic and data operations. Zoho Creator supports real-time collaboration, mobile-friendly access, and seamless integration with other Zoho and third-party services. It's ideal for streamlining business processes like data collection, approvals, and tracking. Whether you're a student, entrepreneur, or enterprise, Zoho Creator empowers you to create powerful applications with minimal effort.

Signing into Zoho Creator

Signing in to Zoho Creator is the first step to accessing its powerful application-building features. To begin, navigate to creator.zoho.com and click on the "Sign In" button. Enter your Zoho credentials, which include your email and password. If you are part of multiple organizations, you will be prompted to select the appropriate one. If you do not have an account, you can easily create one by clicking on "Sign Up." Once signed in, you will be directed to the dashboard, where you can view your existing applications or start creating a new one.

Key Points:

- ◆ Access the platform via a web browser.
- ◆ Requires a Zoho account for sign-in.
- ◆ Dashboard displays existing applications.

Before Creating an Application

Before diving into application creation, it is essential to plan and define your business requirements. This involves identifying the data you need to collect, the workflows you want to automate, and the reports you will require for analysis. Consider the types of users who will interact with the application and their specific needs. Sketching out the layout of your forms and the relationships between different data entities can also be beneficial. This preparatory phase ensures that your application will effectively meet your organization's needs and streamline processes.

Key Points:

- ◆ Define business requirements and data needs.
- ◆ Identify user roles and access levels.
- ◆ Plan workflows and report structures.

Creating a New Application

Once you have a clear plan, you can create a new application in Zoho Creator. Click on the "+ Create Application" button on the dashboard. You will have the option to start "From Scratch" for a fully customized app or choose "From Template" to utilize pre-built solutions tailored for common business needs. After selecting your preferred option, provide a name and description for your application, and choose the workspace or folder where it will be stored. Finally, click "Create" to initiate the application-building process.

Key Points:

- ◆ Choose between starting from scratch or using a template.
- ◆ Provide a name and description for clarity.
- ◆ Select the appropriate workspace for organization.

Forms

Form is used to collect and store data from users. It acts as the main input screen of your application. You can easily create forms using drag-and-drop fields like text, date, dropdown, number, and more. Each form automatically creates a database table behind it to save the submitted data. Forms can also include rules, validations, and subforms for advanced data handling. They are the foundation for creating reports, workflows, and automations in your app.

Features:

- Drag and Drop interface
- Custom fields
- Validation rules
- Subforms
- Default values and Calculations

Example:

For leave approval app the form will have fields like Student name(first name, last name), Roll number, Email, leave type, Reason for leave, start date, end date, approval status etc...

Construction of forms

◆ Planning

1. Fields
2. Sections
3. Validation rules

◆ Creating form

1. Navigate to form
2. Select type
3. Add fields

◆ Configuration

1. Validation
2. Permissions
3. Conditional logic

Reports

A Report in Zoho Creator is used to view and manage the data collected through forms. It displays the submitted data in formats like lists, spreadsheets, calendars, Kanban boards, or charts. Reports help users search, filter, sort, and analyze the stored data easily. You can customize what columns to show and control who can view or edit the data. Reports are directly linked to forms and are essential for tracking and monitoring user inputs. They can also include buttons or custom actions to update or interact with the records. Reports provide dynamic data visualization based on the information collected from forms. They allow users to:

- Filter, sort, and group large datasets.
- Generate statistics (sums, averages, counts).
- Visualize trends using charts (bar, pie, line).
- Export data (PDF, Excel, CSV).

Example:

An "Employee Attendance Report" shows daily check-ins, absentees, and late arrivals in a company.

Pages

Pages in Zoho Creator are custom-designed interfaces that allow users to create dashboards, visual layouts, and interactive elements within an application. Unlike forms and reports, which are more data-driven, pages offer complete design flexibility to present information in a structured, user-friendly manner. Developers can add buttons, charts, panels, embedded forms, and even write scripts to make the interface interactive. Pages are especially useful when you want to create a centralized view or a control panel for users. They enhance the user experience by offering a polished, application-like feel to business solutions.

Key Features:

- ◆ Drag-and-drop builder for effortless UI assembly.
- ◆ Pre-built templates (tabs, grids) to accelerate development.
- ◆ Mobile-responsive design for cross-device accessibility.
- ◆ Role-based permissions to control component visibility.
- ◆ Deluge scripting support for dynamic logic (e.g., conditional displays).
- ◆ Embeddable components (forms, reports, charts) in a single view.
- ◆ Custom HTML/CSS integration for advanced customization.

Steps involved in creating pages:

- ◆ Choose page type
- ◆ Design the page
- ◆ Configure logic
- ◆ Save and Publish

Workflow

Workflows in Zoho Creator are used to automate actions and define business logic within an application. They help trigger specific operations—like sending emails, updating data, or validating inputs—based on user actions such as form submissions, field updates, or schedule-based events. Workflows reduce manual effort and ensure consistent processes across the application. They are built using a combination of visual builders and Deluge scripting, giving both flexibility and control. Overall, workflows are essential for creating smart, responsive, and efficient business applications.

In simple terms we say that A workflow is a sequence of automated steps triggered by an event (e.g., form submission, field update, or scheduled time). It performs actions like sending emails, updating records, or integrating with third-party apps.

Types of workflow

- ◆ Form based workflow
- ◆ Time based workflow
- ◆ Field based workflow
- ◆ Button based workflow

Example:

Leave Approval System

Trigger: Employee submits a leave request.

Condition: If leave days > 3, notify HR.

Actions:

Send approval request to manager.

Update leave balance after approval.

Sync approved leaves to Google Calendar.

Deluge

Deluge is a scripting language used in Zoho Creator to add logic and automation to your apps. It helps you control how your application behaves—like checking if a form field is filled correctly, sending an email when someone submits a form, or updating data automatically. Even if you're not a full-time programmer, Deluge is easy to understand and write. It makes your app smarter by allowing you to create custom actions based on user input or system events.

Syntax of deluge:

For finding voting eligibility

```
name = input.Name;
    if (input.Age >= 18)
    {
        info "You are eligible.";
    }
    else
    {
        info "You are not eligible.";
    }
```

I intend to integrate the cybersecurity technique of data masking into this project to enhance the security of sensitive information.

Project - Secure AI-Driven Loan Evaluation System Using Zoho Creator

Project Objective

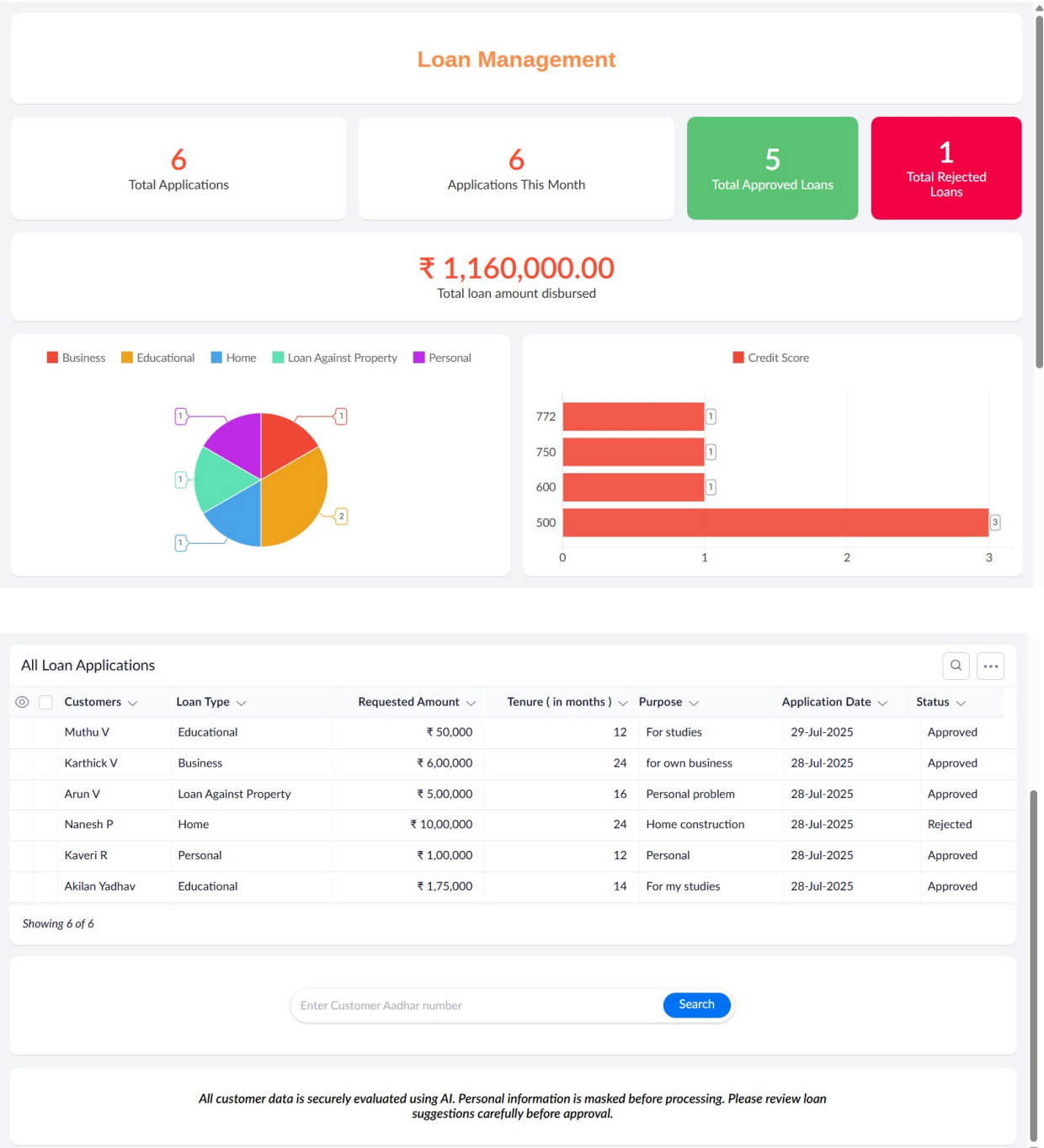
The objective of this project is to develop a secure, efficient, and intelligent loan evaluation system using Zoho Creator that integrates Artificial Intelligence (AI) for credit analysis. The system aims to protect sensitive applicant data through masking techniques while utilizing ChatGPT to evaluate loan eligibility based on the applicant's input. This project combines low-code application development, AI integration, and cyber security principles to enhance the decision-making process for loan approvals.

Project Description

This project focuses on building a secure and intelligent loan application management platform using Zoho Creator, a low-code development environment. The core innovation lies in the integration of ChatGPT, an AI-powered language model, to assist in evaluating the creditworthiness of loan applicants. To maintain data privacy and security, sensitive applicant data—such as names, contact details, PAN numbers, and income figures—is masked before being sent to the AI for analysis. The AI processes this masked data and returns an eligibility result in a similarly anonymized format. The system then maps the AI-generated result back to the actual applicant record securely, enabling administrators to make informed decisions without exposing private data during processing.

The application includes features such as applicant data entry forms, credit score validation, document verification status, and an admin dashboard for decision-making. Deluge scripting is used to implement logic such as, automated workflows, and secure data handling. By combining AI-driven analysis with strong cybersecurity practices, this project ensures both intelligent processing and protection of user information.

1. Dashboard



2.Adding customers

customers

Full Name

First Name

Last Name

Mobile Number

 +91

Email ID

Date of Birth

Aadhar Number *

PAN Number *

Address

Address Line 1

Address Line 2

City / District

State / Province

-Select-

▼

Postal Code

Country

Employment Type

-Select-

▼

Annual Income

Submit

Reset

All Customers

☐	Full Name ▼	Mobile Number ▼	Email ID ▼	Date of ... ▼	Aadhar Nu... ▼	Address ▼	Employr
☐	Muthu V	 +919461496412	muthu@gmail.com	01-Mar-2000	456123567854	 133/1.ascdjhm, hajllsjdmd, dgvgecekc, Tamil Nadu, 608302, India	Governm
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☐	Arun V	 +919856231145	arun@gmail.com	10-Aug-2005	789465234523	 144, namma street, chidambaram, Tamil Nadu, 608001, India	Agricultu
☐	Karthick V	 +917777777777	vikram@gmail.com	07-Mar-2000	895526256632	 56, edham nagar, cuddalore, Tamil Nadu, 604523, India	Private

Final Output

Loan Applications

Customers

Arun V

Loan Type

Educational

Requested Amount

200000

Tenure (in months)

12

Credit Score

750

Purpose

For my son studies

Application Date

28-Jul-2025

Chatgpt Input

Based on the following customer details, please analyse and determine whether the applicant is eligible for a loan.

Analysis

☐ Chatgpt

Evaluation Comments

Based on the customer details provided, the applicant named Arun appears to be eligible for the loan. Here are the reasons:

1. PAN number and date of birth are valid identification details.
2. Annual income of 200,000 meets the requirement for the requested loan amount of 200,000.
3. Employment type in agriculture shows a stable source of income.
4. The loan type is educational, which suggests a legitimate purpose for borrowing.
5. The requested loan amount and tenure seem reasonable based on the applicant's income.
6. A credit score of 750 indicates good creditworthiness and financial discipline.

Overall, Arun meets the eligibility criteria for the loan based on the information provided.

On the above image a loan evaluation prompt was provided to the ChatGPT API using customer details. The AI analyzed the data and generated a clear eligibility decision along with reasoning in the evaluation comments.

Recommended Amount

200,000

Status

- ☐ Submitted
- ☐ Under Review
- ☒ Approved
- ☐ Rejected

Submit

Reset

After the ChatGPT analysis, the admin reviews the evaluation, sets the recommended loan amount, selects the appropriate application status, and submits the final decision

Example customer detail after analysing

Customers	Nanesh P
Loan Type	Home
Requested Amount	₹ 10,00,000
Tenure (in months)	24
Purpose	Home construction
Application Date	28-Jul-2025
Credit Score	600
Evaluation Comments	Based on the customer details provided, it is recommended not to give a loan as the annual income of 840,000 is lower than the requested loan amount of 1,000,000. The exact recommended amount would be based on the customer's annual income and credit score, which may not meet the criteria for the requested loan amount.
Recommended Amount	
Status	Rejected

 [Add a comment](#)

Learning Reflection

During the course of this project, I had the opportunity to gain hands-on experience in building a real-world business application using Zoho Creator, a powerful low-code development platform. As a first-year student, this experience helped me understand how digital workflows, forms, reports, and Deluge scripting can be used to automate and streamline business operations such as loan processing.

Additionally, I explored core cyber security concepts like data masking, anonymization, and encryption, and learned how they can be applied practically to protect sensitive information in web applications. Integrating AI technology via ChatGPT API introduced me to the potential of intelligent systems in assisting with decision-making, while also emphasizing the importance of data privacy when working with third-party APIs.

This project has strengthened both my technical skills and my understanding of how software development and cyber security principles come together in solving real-world business challenges.

Conclusion

This project demonstrates the integration of artificial intelligence and cyber security within a low-code business application environment using Zoho Creator. By leveraging the capabilities of the ChatGPT API, the system effectively analyzes customer data and generates accurate loan evaluations. The incorporation of data masking ensures the protection of personally identifiable information during the evaluation process, maintaining both compliance and trust. Through structured form design, automated workflows, and intelligent decision-making, this solution simplifies the loan assessment process for administrators. Overall, this project highlights the potential of combining AI and secure development practices to build smarter, safer, and more efficient business applications.